Video Transcript

Video 1: Introduction (04:51)

So, this week we're going to continue our quest to understand machine learning and we're going to cover two techniques, reinforcement learning and also, deep neural nets. We're going to go into some depth with reinforcement learning, partly because it's one of my favorite methods and it was used by the AlphaGo team in Google to beat the world champion at Go and also the world champion at chess.

So, we're going to apply it to a robot that needs to find its way around. Now, in any of these situations, we need to build a map of where we are. So, even in, say, Go, we need a map of the state of the board. Now, in our case, we're going to have a robot that's finding its way through rooms, and we can only go if we're in room one, we can only go to certain other rooms, maybe we can go to two or to three, but we can't go to five or to six. So, one of the first things we need to understand is how we build up a map of our environment.

And the next thing is, we need to have a strategy for exploring. So, in the case of rooms, we have a number of, what we call, states. So, a state would be what room we're in. And then given that we're in a certain room, there are only certain actions that we can take. So, as I said, if we're in room one, we may be able to go to room two, but we can't go to room five. So, the first thing is we need to lay out the environment and have a way of expressing it.

So, we're going to express the rooms as a matrix. And, given that we're in a state, so we've got a matrix of states. And then at every state, we have a certain number of actions that are possible. So, if we're in room one, we can move to room two, but we can't move to rooms four, or five, or six, etc. Now, the robot needs to find its way to a goal. So, we need some way of expressing a goal and we need to explore. And so, we need to understand how we leave a trail, so that when we're successful, we need to somehow express that success.

So, we're going to have what we call a robot matrix. And if we get to our goal, we're going to have a big robot. Now, we need to understand the quality of the state that we're in at any point. So, if we're close to our goal, we'll have a high quality, we call it a Q matrix. So, the value will be

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higher than if we're a long way from the goal, where we should have a lower quality. So, we're going to go through all of the techniques that machine learning for reinforcement learning uses. And once we've understood that, we're going to apply it to what's called the Frozen Lake problem. So, this is where a robot needs to find its way across a lake, but there are holes in the lake where the ice is melted, and if it goes there, it's going to fall in. So, we'll have fun with the Frozen Lake problem.

And then we're going to move to deep learning and we're going to use Colab. And, Colab is Google's version of Jupyter Notebooks, and we're going to use that to operate in the cloud. We're going to be using machines in the cloud via Colab to solve our problem. And in the deep neural net case, we're going to be doing a problem called Deep Dream, where we input an image of our self and we look at what the various layers in a deep neural net are doing to that image. So, we'll have kind of fun looking at how we're seen by the deep neural net. So, let's go and take a look at reinforcement learning and deep neural Nets. Bye for now.

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